FACULTY OF COMPUTING

UNIVERSITI TEKNOLOGI MALAYSIA

NETWORK COMMUNICATIONS

SECR1213-12

SEMESTER 1 2024/2025

PROJECT TASK 4: MAKING THE CONNECTIONS – LAN and WAN

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GROUP 5

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4.1 Identify the work areas on floor plan

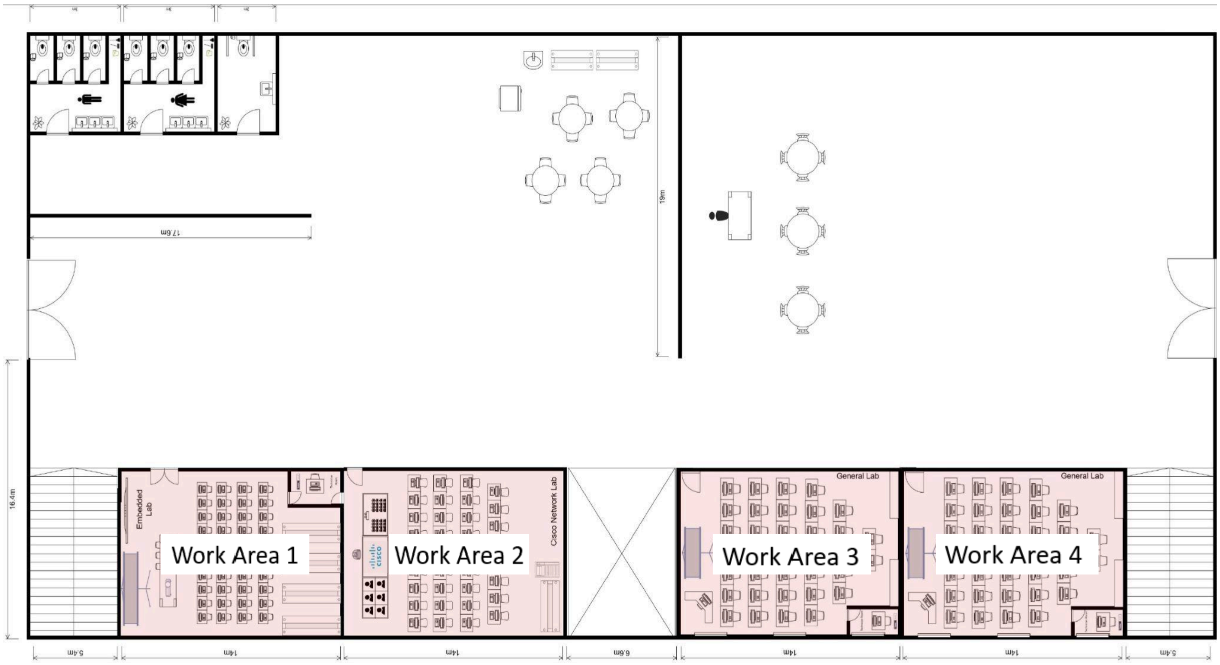


Figure 4.1.1: Work Area at Ground Floor

Work Area 1 shown in figure 4.1.1 is an embedded lab that focuses on IoT and embedded systems, providing a platform for students to work on projects involving sensors, microcontrollers, and other IoT devices. The lab has 32 PCs for students to use as well as IoT devices and sensors that are linked to a PoE-managed switch, which supplies both power and data communication. Wireless access points (WAPs) facilitate the networking of mobile IoT devices, resulting in seamless network integration. This lab is critical for encouraging innovation and allowing students to investigate IoT technologies in a hands-on setting.

Work Area 2 shown in figure 4.1.1 is a Cisco Network Lab with 14x10m size. It is designed to give students hands-on experience with networking technology. It is outfitted with 30 computers to promote interactive learning and practical exercises. The lab is equipped with routers and switches for network simulation and configuration, as well as a 52-port managed switch that supports VLANs for network segmentation. A smartboard and webcams are among the other capabilities that allow for interactive education and remote collaboration.

Work Area 3 and 4 shown in figure 4.1.1 are both general labs that are designed to accommodate a wide range of computing applications, including programming, software development, and data analysis. Each lab has 30 PCs placed in rows to provide students with plenty of workspace. These labs are also outfitted with projectors to aid teaching activities. Their major aim is to provide generic computing environments in which students may complete coursework, collaborate on projects, and improve their technical abilities.

Figure 4.1.2: Work Area at First Floor

Work Area 5 in figure 4.1.2 is a Video Conferencing Room which is designed to use for virtual meetings and remote collaboration. It is outfitted with high-definition cameras, microphones, and smart TV to ensure a professional and immersive conference experience. A dedicated managed switch with Quality of Service (QoS) settings prioritizes video traffic, resulting in seamless and uninterrupted communication. The room's goal is to facilitate smooth virtual interactions for academic, administrative, and collaborative purposes.

Work Area 6 in figure 4.1.2 is a server room. It is the foundation of the building's network infrastructure. It houses the basic network switches, routers, servers, and security appliances such as firewalls and intrusion prevention systems. It is designed to manage and distribute network resources, guaranteeing consistent connectivity and data security throughout the building.

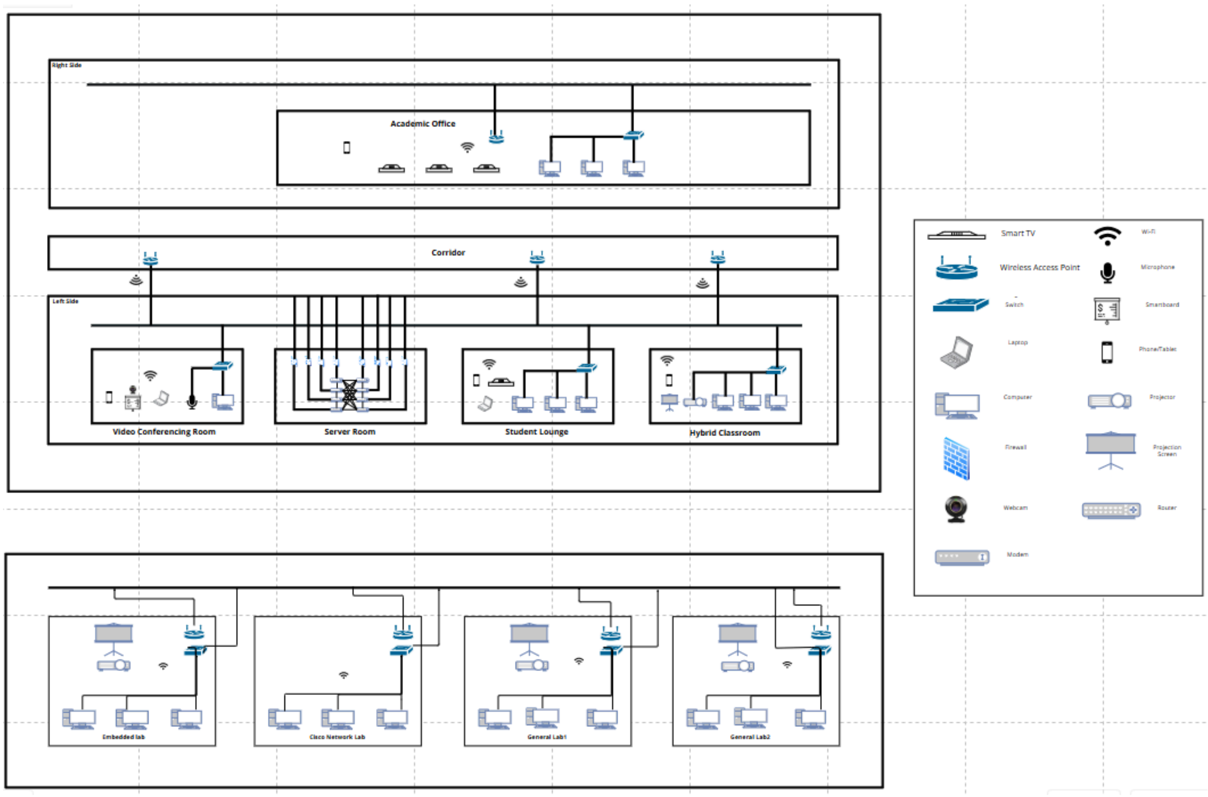
Work Area 7 in figure 4.1.2 is a student lounge. It provides a comfortable environment for informal learning, collaboration, and relaxation. It has comfortable seating, charging stations, and several Wi-Fi 6 wireless access points to support high-density networking. The student lounge encourages casual browsing, online collaboration, and media streaming, giving students a comfortable place to rest and engage.

Work Area 8 in figure 4.1.2 is a hybrid classroom. It is a flexible learning environment designed to accommodate a variety of traditional and modern teaching approaches. It has 30 computers set up to allow for both individual work and group collaboration. A 52-port controlled switch manages wired connections, and wireless access points provide connectivity for personal devices. The room has a projector and a lecturer's workstation, allowing teachers and students to participate in interactive learning sessions. The major goal of this classroom is to accommodate varied teaching techniques while also enhancing the learning experience through modern technology.

Work Area 9 in figure 4.1.2 is an academic office and there are two staff meeting rooms. It is designed for administrative work, meetings, and collaborative projects. These rooms are outfitted with smart TV and 21 workstations. A Wi-Fi 6 wireless access points is installed to ensure wireless connectivity. These rooms serve as hubs for academic and administrative processes, providing a peaceful and connected setting for employees to work effectively.

4.2 The distribution and connection of devices

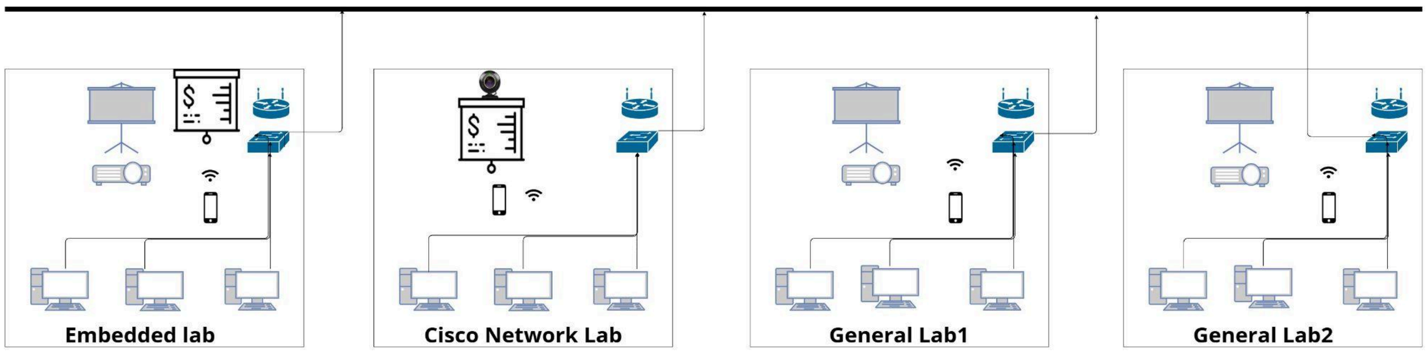
4.2.1 Network Distribution Diagram



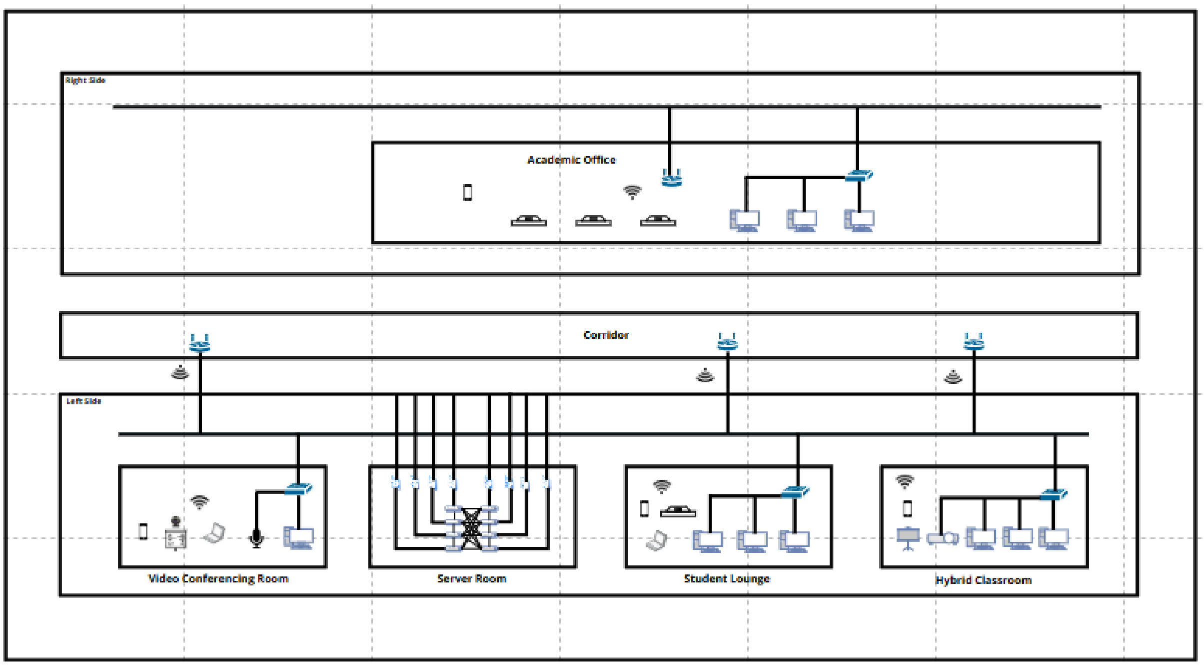
4.2.1.1 Network Distribution Diagram

Figure 4.2.1.1 shows the detailed layout of the distribution of the network within the Faculty of Computing, including the setup and connections between the routers, switches, wireless access points, end-user devices, and cable paths showing the logical and physical topologies.

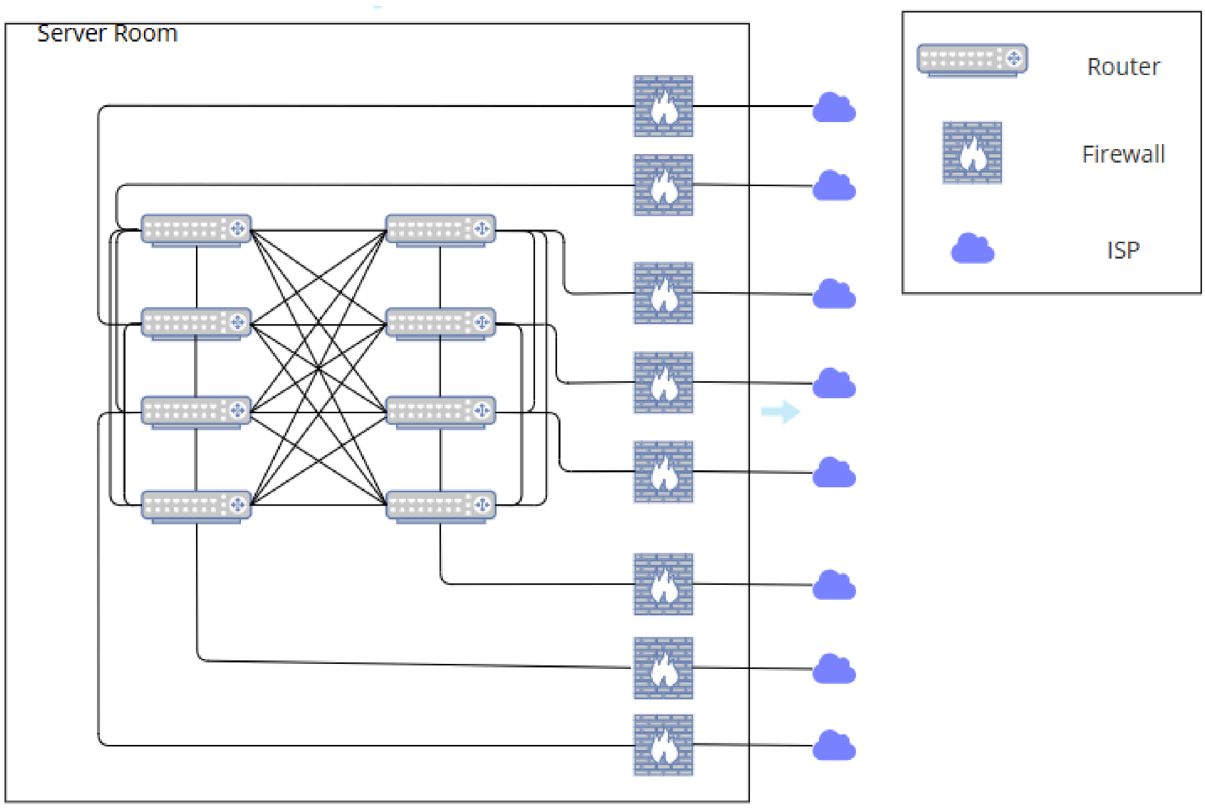
Figures 4.2.1.2 focus on the ground floor, where two general labs, one embedded and a Cisco networking lab are located. Figure 4.2.1.3 highlights the first floor where the Academic Office, Video Conferencing Room, Server Room, Student Longue and Hybrid Classroom. The above workspaces are connected to a router that connects to a modem using Ethernet cables, thereby offering better connectivity and performance compared to wireless options. The modem then connects to a wireless access point using fiber optic cables, which enables the wireless connectivity of end devices like laptops and mobile phones so that they can access the Internet using assigned IP addresses.



4.2.1.2 Network Distribution Diagram-Ground Floor



4.2.1.3 Network Distribution Diagram-First Floor



4.2.1.3 Network Distribution Diagram Server Room

The above diagram is the network distribution for the Server Room. It consists of a series of interconnected switches in a mesh topology that provides redundancy, and it ensures the high availability of internal communication between them. Other than that, these switches are connected individually with firewalls that act as security gateways, controlling and securing traffic between the internal network and external connections. Nonetheless, each firewall is linked to a single router, responsible for routing data packets between the internal network and external networks (the Internet). Lastly, the routers are connected with Internet Service Providers (ISPs) or cloud services to enable external communication.

4.2.2 Room Level Network Distribution Diagram

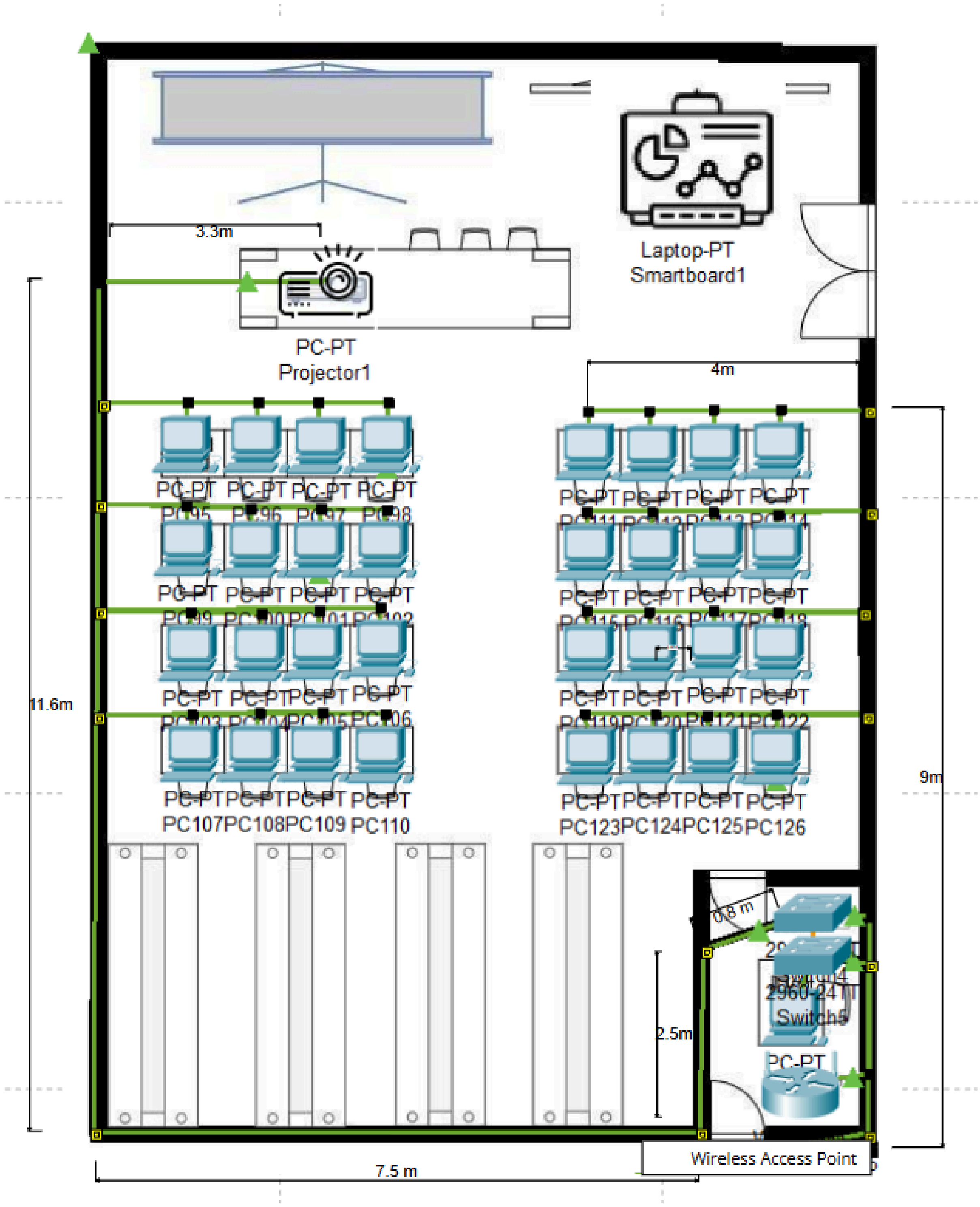


Figure 4.2.2.1: Embedded Lab

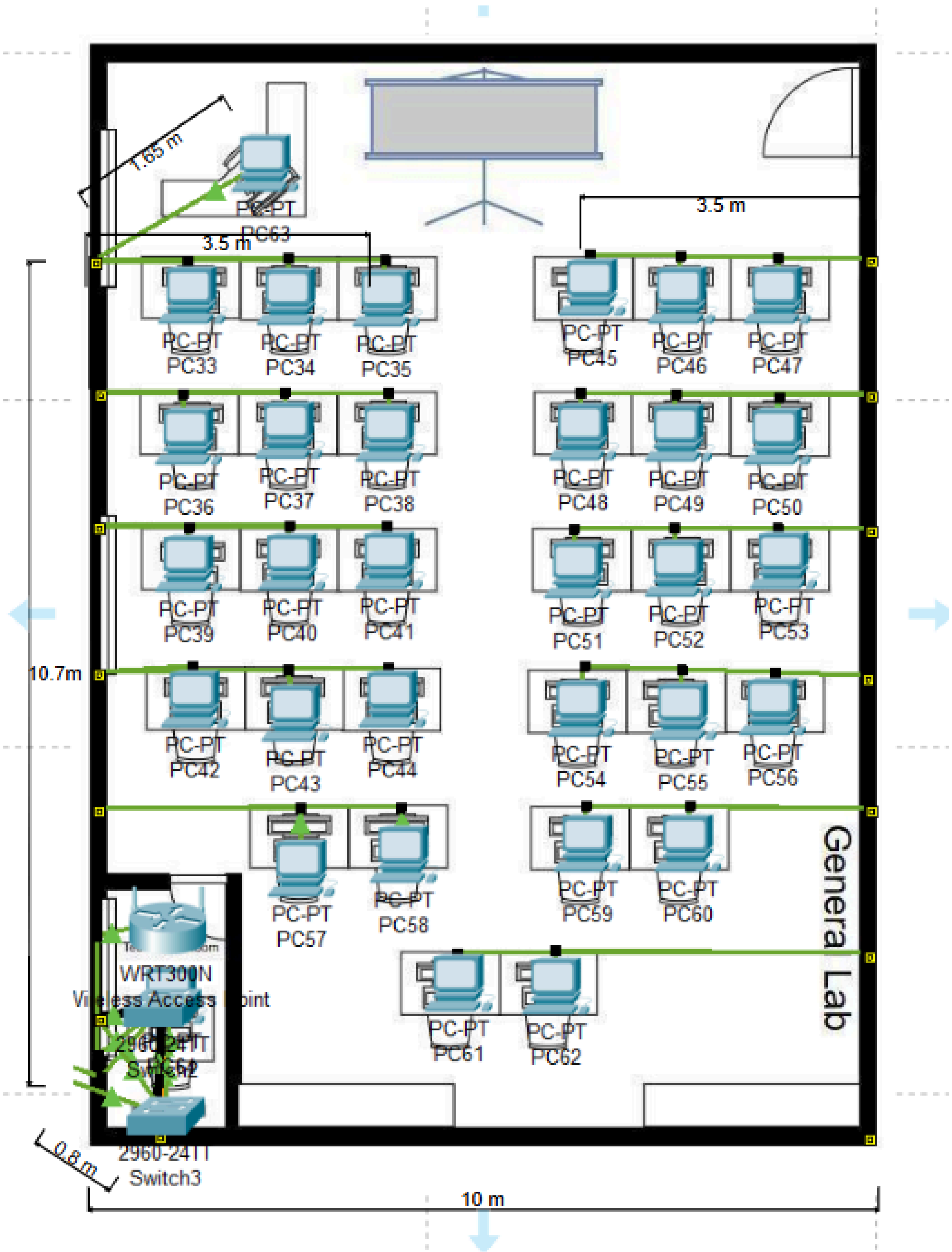


Figure 4.2.2.2: General Lab

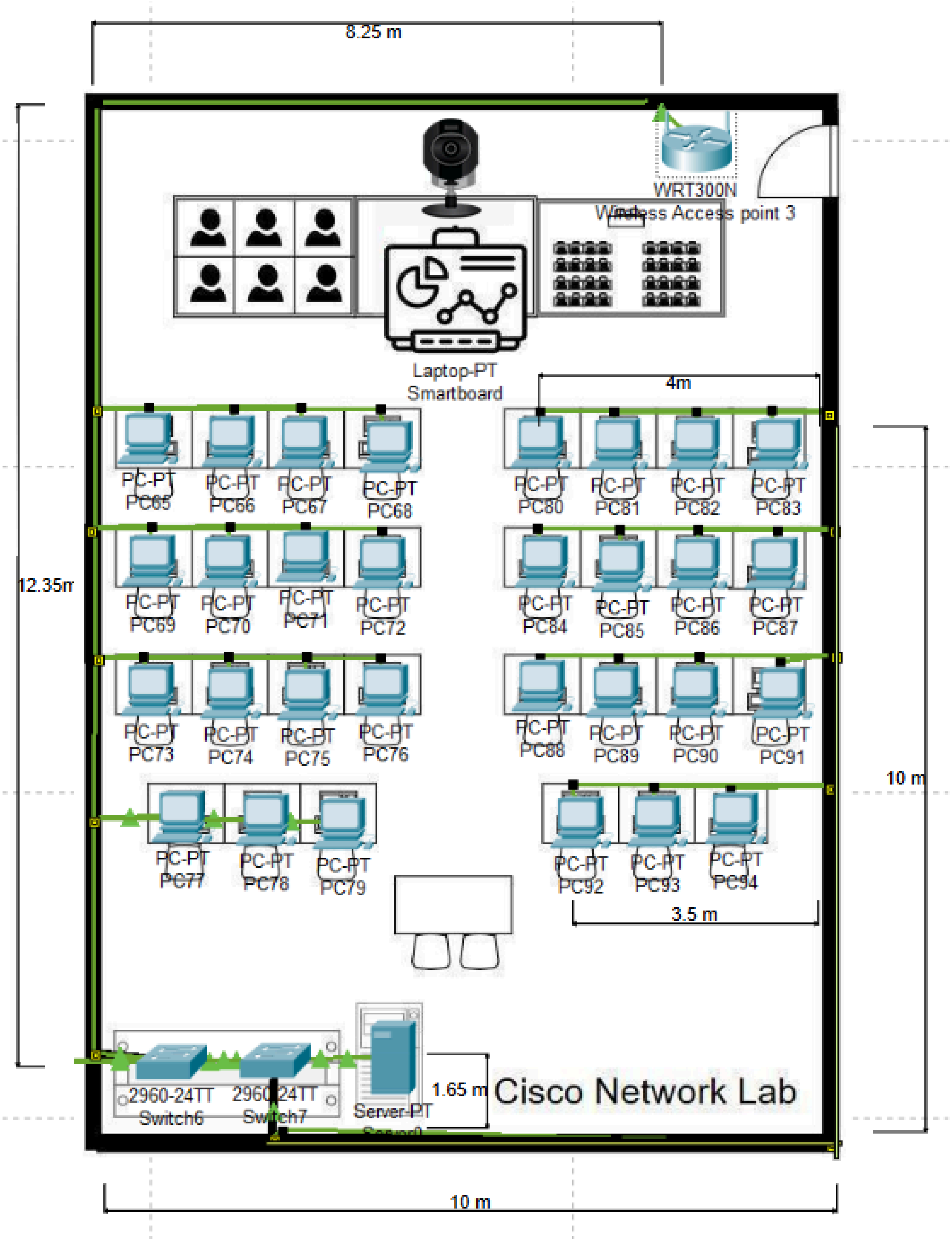


Figure 4.2.2.3: Cisco Network Lab

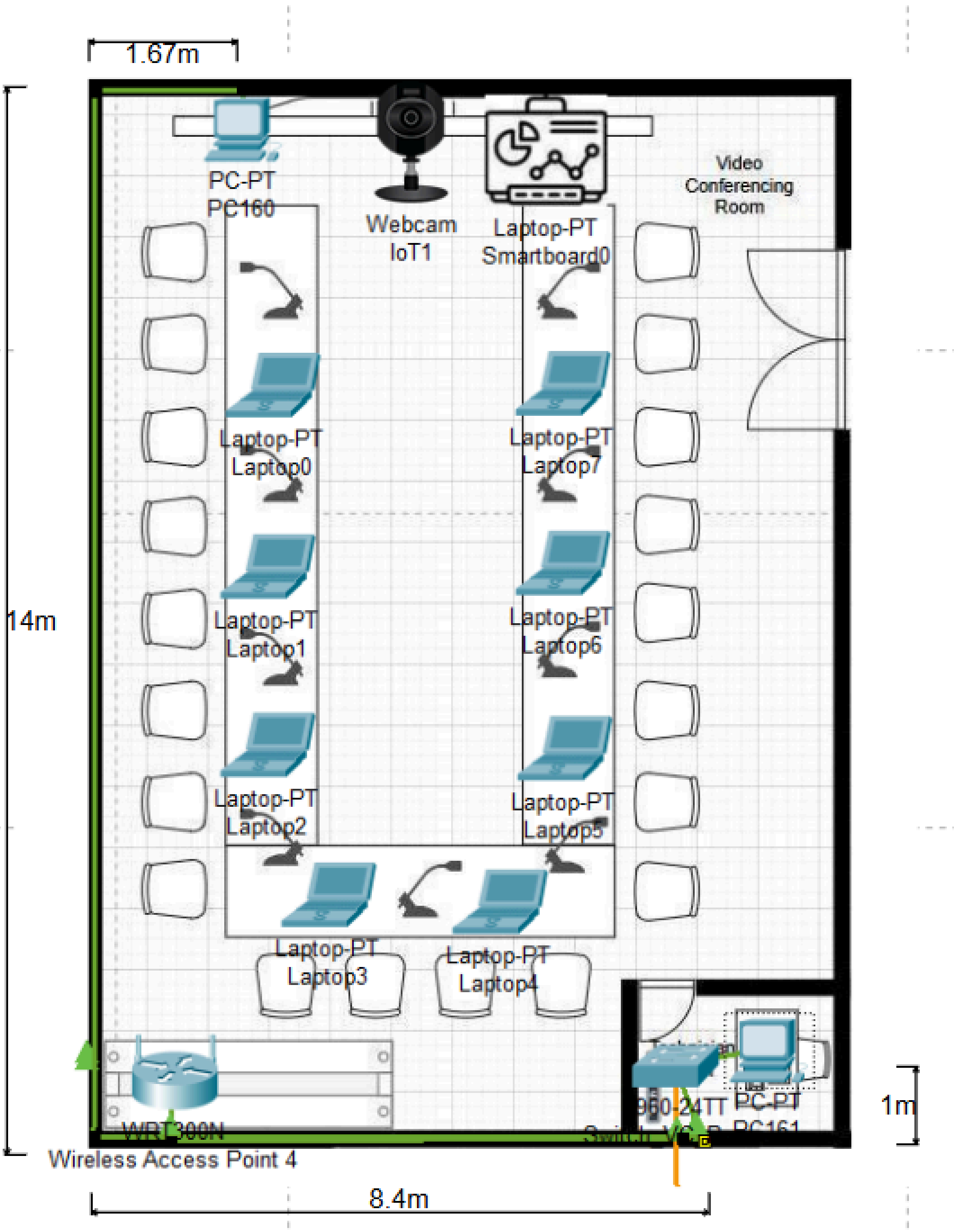


Figure 4.2.2.4: Video Conferencing Room

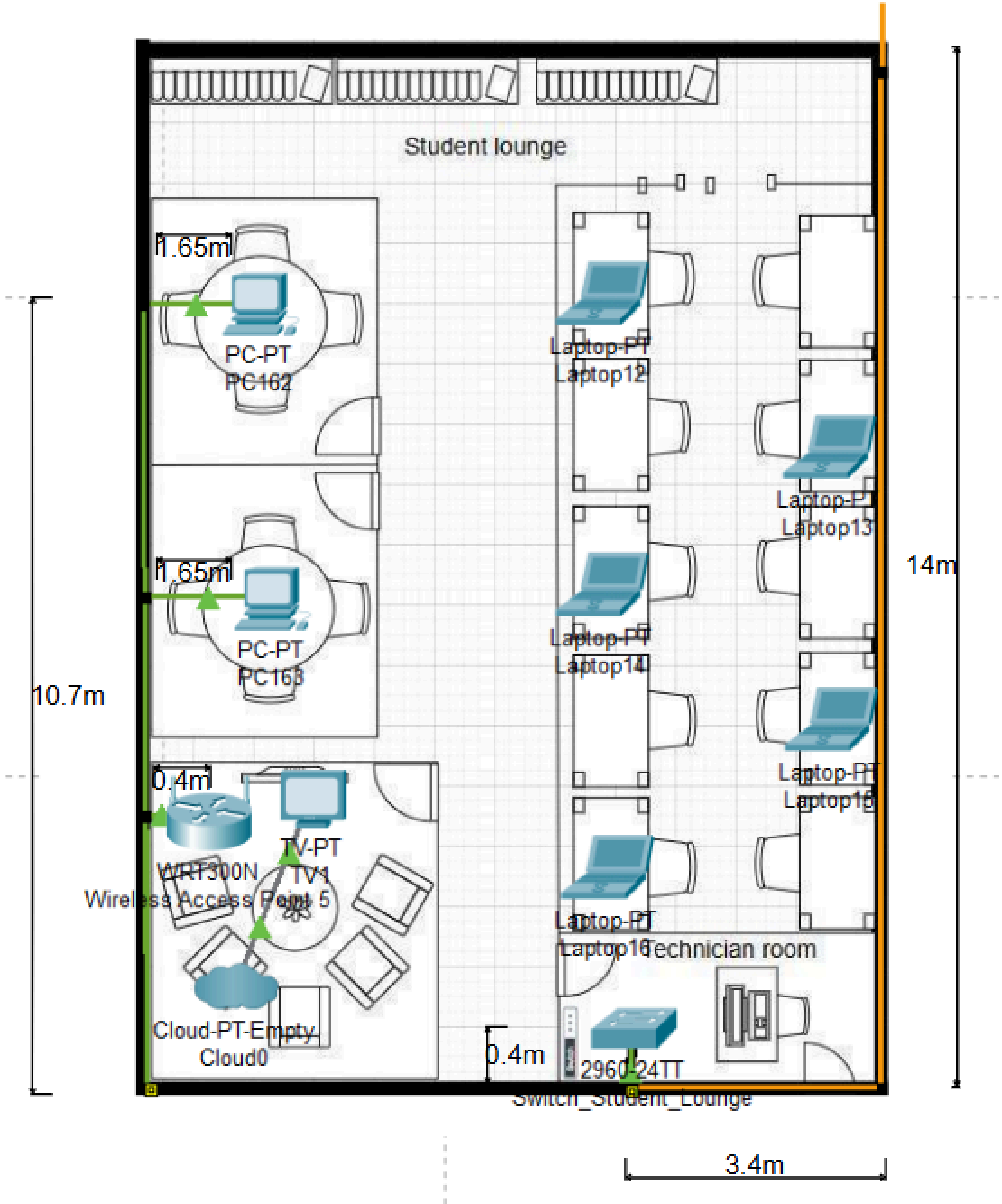


Figure 4.2.2.5: Student Lounge

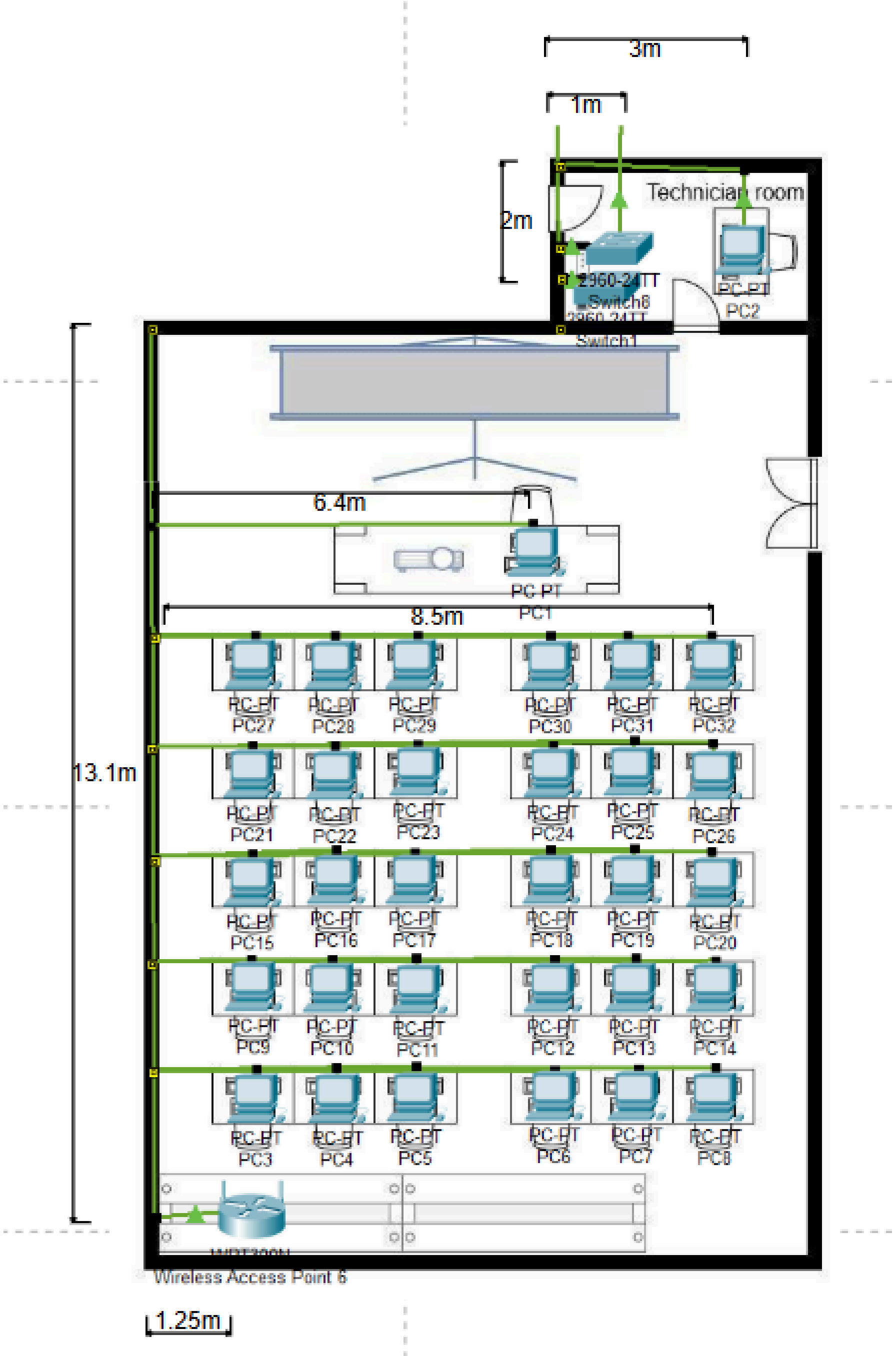
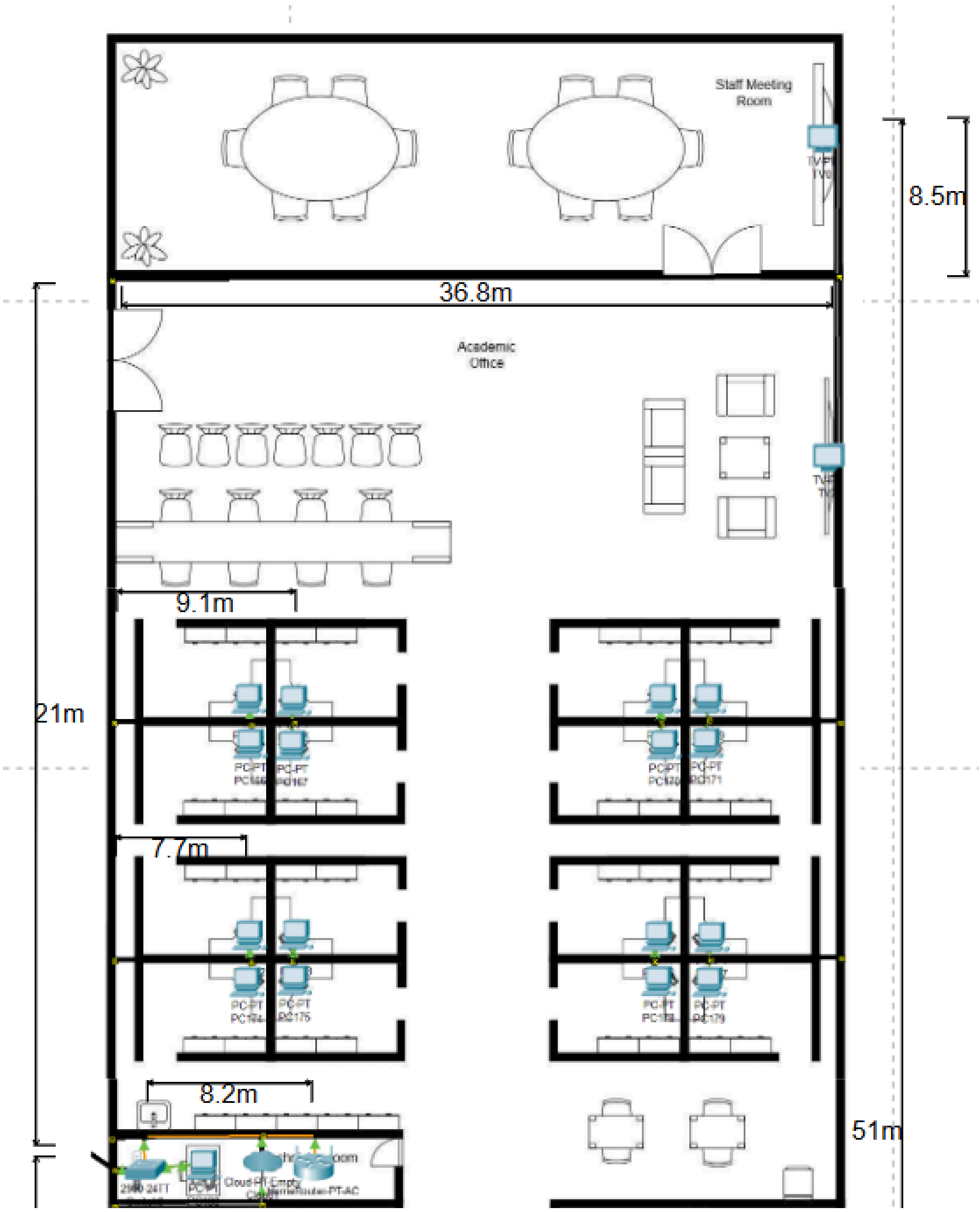


Figure 4.2.2.6: Hybrid Classroom



Staff Meeting Room
8.5m
36.8m
Academic Office
9.1m
21m
7.7m
8.2m
51m

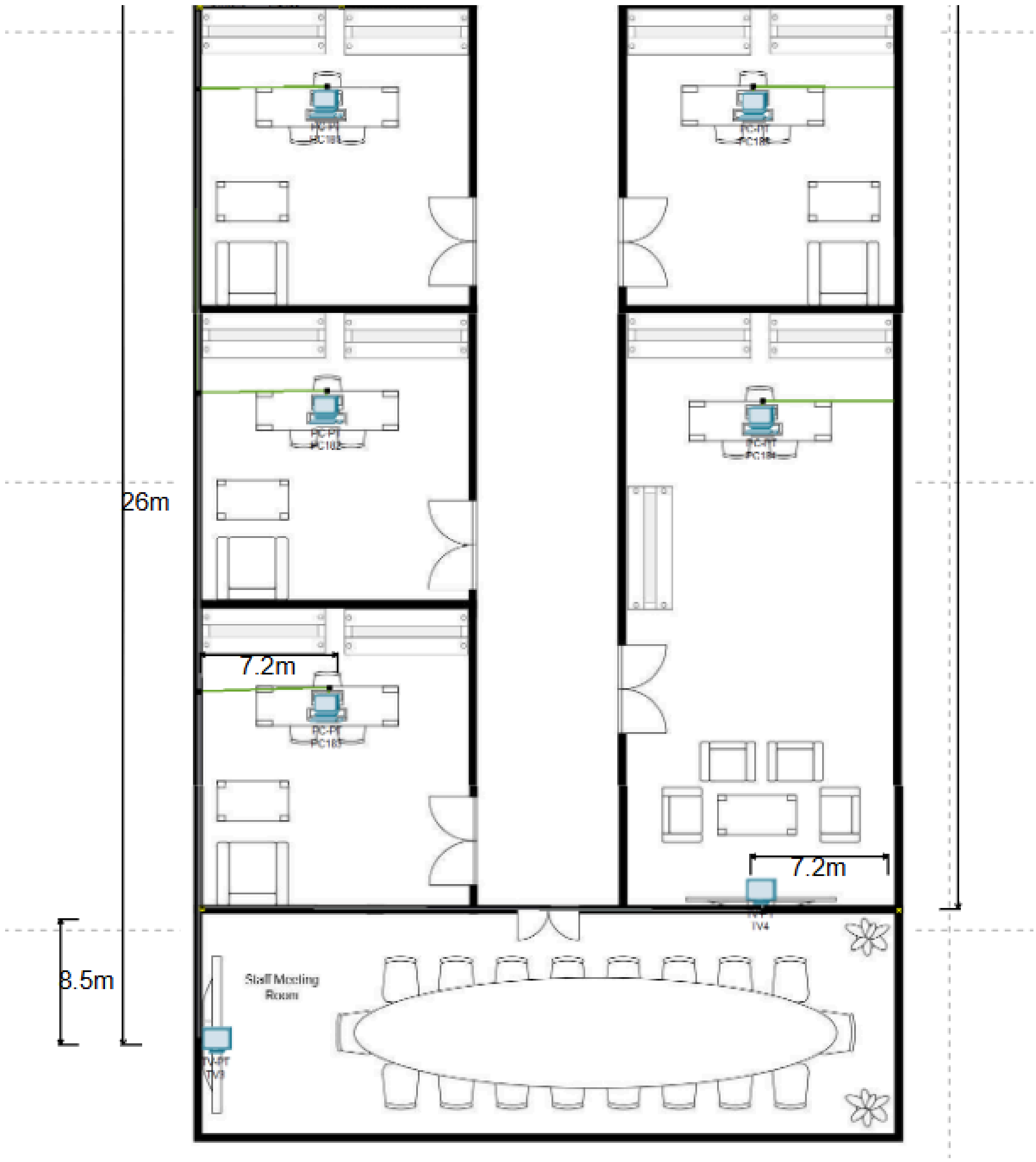


Figure 4.2.2.7: Academic Office

4.2.3 Floor Diagram

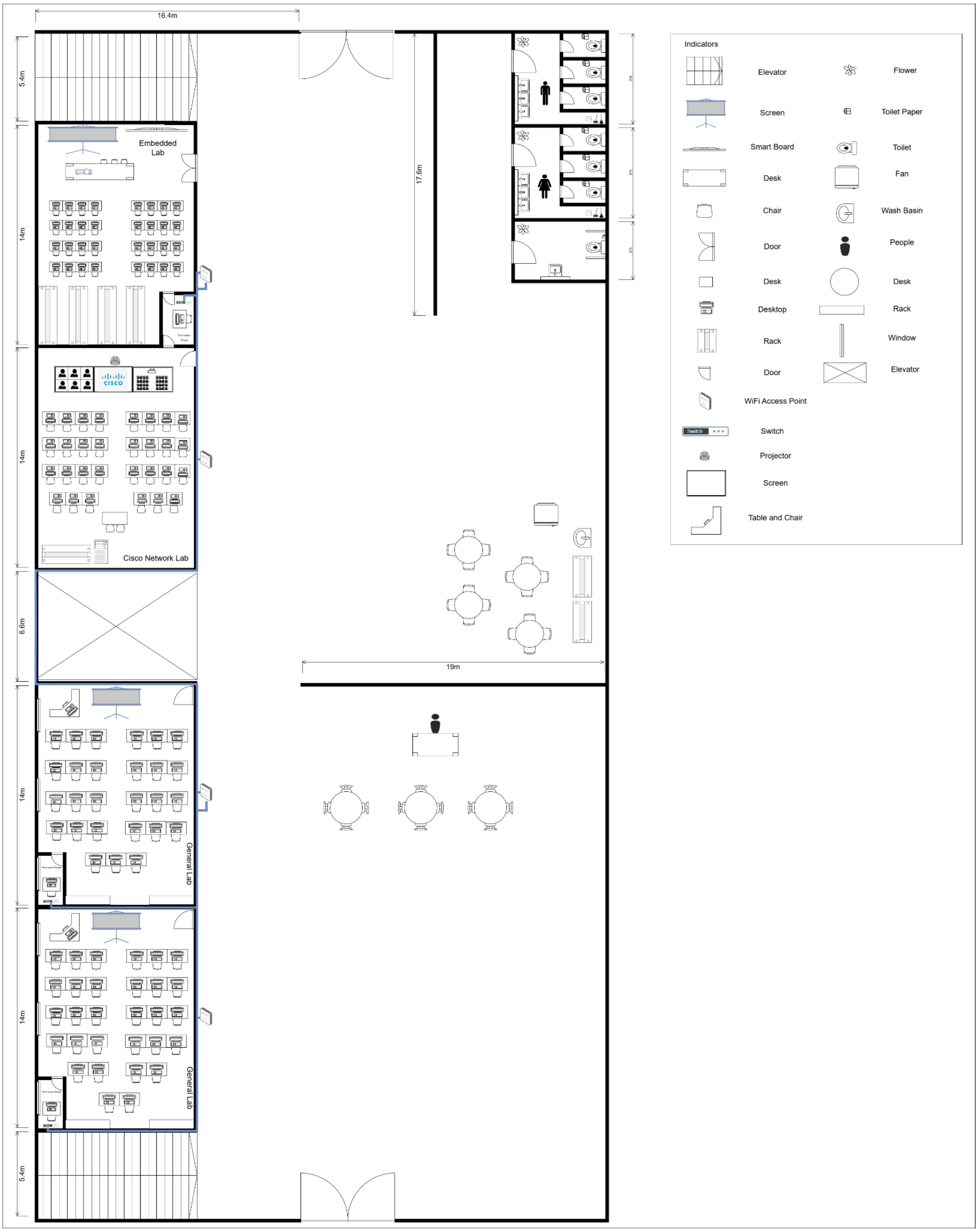


Figure 4.2.3.1: First Floor

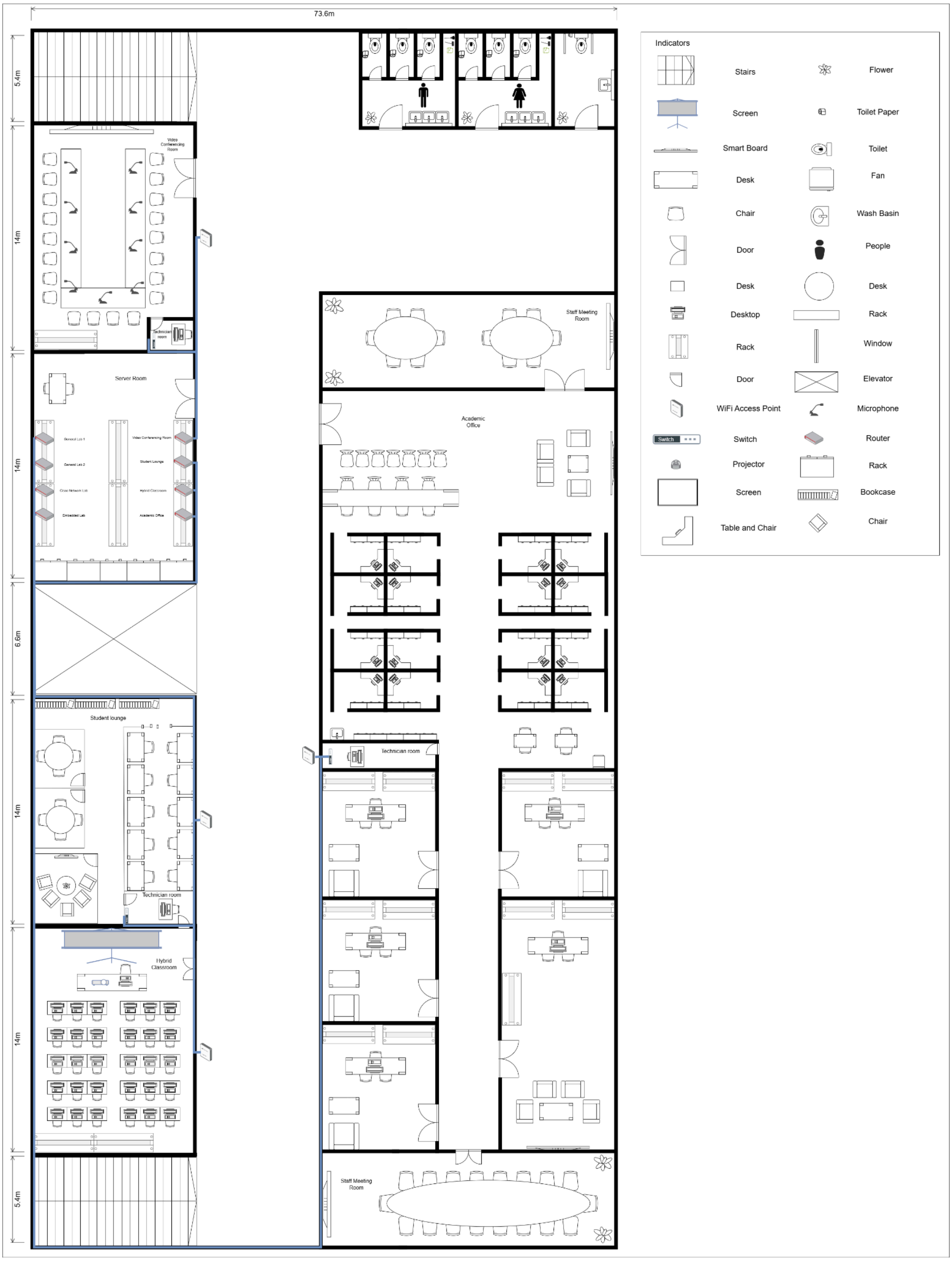


Figure 4.2.3.1: Second Floor

4.3 Identify number of connections, patch cords and switch ports

Area	**Number of Switch Port Needed**
Embedded Lab	35
Cisco Network Lab	31
General Lab 1	33
General Lab 2	33
Video Conferencing Room	3
Server Room	4
Student Lounge	5
Academic Office	27
Total	141

Table 4.3.1: Number of Switch Ports Needed

Area	**Number of Patch Port Needed**
Embedded Lab	35
Cisco Network Lab	31
General Lab 1	33
General Lab 2	33
Video Conferencing Room	3
Server Room	4
Student Lounge	5
Academic Office	27
Total	141

Table 4.3.1: Number of Patch Ports Needed

Area	**Number of RJ45 Connector Needed**
Embedded Lab	70

Cisco Network Lab	62
General Lab 1	66
General Lab 2	66
Video Conferencing Room	6
Server Room	8
Student Lounge	10
Academic Office	54
Total	282

Table 4.3.1: Number of RJ45 Connector Needed

4.4 Identify cable types and length

Description	Cable Type	Length(m)
Ground Floor		
Embedded Lab	CAT 6 cable	63.4
Cisco Network Lab	CAT 6 cable	73.25
General Purpose Lab 1	CAT 6 cable	74.15
General Purpose Lab 2	CAT 6 cable	74.15
Server Room to Embedded Lab	Fiber Optic Cable	70
Server Room to Cisco Network Lab	Fiber Optic Cable	80
Server Room to General Purpose Lab 1	Fiber Optic Cable	75
Server Room to General Purpose Lab 2	Fiber Optic Cable	65
The total length of the Ground Floor (m)		574.95

Description	Cable Type	Length(m)
First Floor		
Video Conferencing Room	CAT 6 cable	25.10
Student Longue	CAT 6 cable	32.20
Hybrid Classroom	CAT 6 cable	70.25
Staff Meeting Room1	CAT 6 cable	8.50
Academic Office	CAT 6 cable	202.80
Staff Meeting Room2	CAT 6 cable	8.50
Server Room to Hybrid Classroom	Fiber Optic Cable	27.6
Server Room to Student	Fiber Optic Cable	13.60

Longue		
Server Room to Video Conferencing Room	Fiber Optic Cable	7.00
Server Room to Academic Office	Fiber Optic Cable	36.80
Server Room to Staff Meeting Room1	Fiber Optic Cable	42.30
Server Room to Staff Meeting Room2	Fiber Optic Cable	70.00
The total length of the First Floor (m)		544.65

In the network infrastructure of the Faculty of Computing building, **Cat6 cables** and **fiber optic cables** are used to establish reliable and high-performance connections between devices, switches, and the core network in the server room.

Cat6 cables connect end-user devices like PCs, projectors, IoT devices, smartboards, and wireless access points (WAPs) to the nearest access switches. These cables are designed for short-distance connections and can enable high-speed data transfer rates of up to 1 Gbps over distances of up to 100 meters. Cat6 cables connect 30 PCs and extra devices in each lab to their respective access switches, offering efficient and dependable wired connectivity. Similarly, in the General Purpose Labs and Hybrid Classroom, Cat6 cables connect all devices to access switches, ensuring adequate bandwidth for general computer applications and instructional purposes. These cables are the backbone for local connections within rooms or work areas.

Fiber optic cables serve as backbone connections, connecting access switches in labs, classrooms, and other work areas to the main router in the server room (Main Distribution Frame MDF). Fiber optic cables are preferred because they can transport data over long distances without signal loss and at significantly faster speeds than Cat6 cables. Switches in the Cisco Network Lab, Embedded Lab, General Purpose Labs, and other areas are connected to the router in the server room by fiber optic cables, allowing for high-speed connectivity and centralized network control. These cables also offer scalability and future-proofing because they can manage increasing bandwidth requirements, making them perfect for a modern and expanding network architecture. The use of fiber optic cables between access switches and the core router ensures minimal latency and high performance for crucial network operations.

In conclusion, by combining Cat6 cables for device-level connections and fiber optic cables for linking switches to the core router in the server room, this network infrastructure ensures seamless connectivity and high performance across all work areas in the building.

Meeting Minutes

Date/Time	7 Jan 2025 3:00pm
Location	Online (Google Meet)
Agenda	1. Discuss the question on Task 4
	2. Discuss report format
	3. Task Separation and Distribution
Meeting Host	Evelyn Goh Yuan Qi

Attendance		
Name	**Time**	**Reason For Absence**
Goh Jiale	3:00pm	-
Joanne Ching Yin Xuan	3:00pm	-
Evelyn Goh Yuan Qi	3:00pm	-
Lim Yu Han	3:00pm	-

Minutes			
No.	**Item Discussed**	**Details**	**Person-In-Charge**
1	Discuss of question on Task 4	Joanne suggests using Visual Paradigm to complete the network distribution diagram.	Joanne
		Evelyn suggests that the floor diagram can use the original diagram in task 1 that draws in Cisco Packet Tracer.	Evelyn
2	Discuss Report Format	Goh Jiale and Lim Yu Han list down the deliverables for task 4	Goh Jiale Lim Yu Han
3	Task Distribution	Identifies number of connections + Network Distribution Diagram for 2nd Floor and room level	Yuhan
		Identifies cable types and length + Network Distribution Diagram for 1st Floor and room level	Joanne

		Identifies the work areas on floor plan + Identifies cable types and length	Evelyn
		Identifies number of connections + Network Distribution Diagram Server Room	Goh Jiale
4	Meeting ended	Lim Yu Han will do the next meeting minute.	Lim Yu Han
		The next meeting is on 15th January at 2pm through Google Meet (online).	All Member